

- 1 A golfer is practising his tee shot from a platform 7.0 m off the ground as shown in Fig. 1.1. The golf ball was launched at a speed of 50 m s^{-1} , 40° above the horizontal. Assume air resistance is negligible.

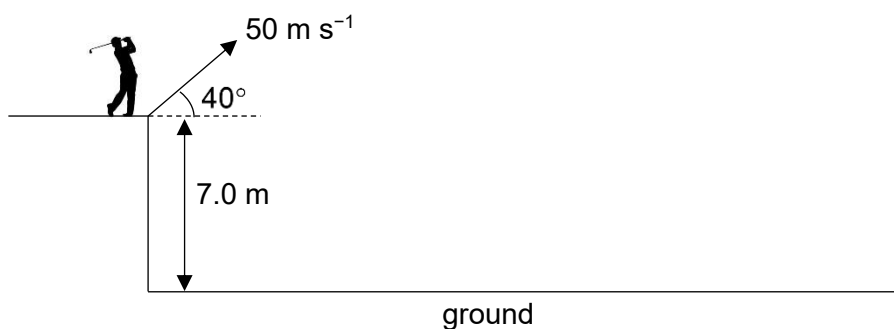


Fig 1.1

- (a) Determine the maximum height above the ground attained by the ball.

Initial vertical velocity $u_y = 50 \sin 40^\circ = 32.139 \text{ m s}^{-1}$ [1]

At the top of flight the vertical velocity $v_y = 0 \text{ m s}^{-1}$

Using $v_y^2 = u_y^2 + 2as$, taking up as positive, we get

$$0 = 32.139^2 + 2(-9.81)s$$

$$s = 52.647 \text{ m}$$

$$\text{Max height} = 52.647 + 7.0 = 59.647 = 60 \text{ m}$$

Can also solve via energy considerations

maximum height = m [3]

- (b) Calculate the time of flight of the ball.

When ball touches the ground $s_y = -7.0 \text{ m}$

Using $s_y = u_y t + \frac{1}{2} a t^2$,

$$-7 = 32.139t + \frac{1}{2}(-9.81)t^2$$
 [1]

$$t = -0.211 \text{ s (reject) or } t = 6.7639 = 6.8 \text{ s}$$
 [1]

time of flight = s [2]

- (c) A golf ball typically bounces a few times after a tee shot as shown in Fig. 1.2. The first time the ball touches the ground is indicated by A and the fourth time it touches the ground is indicated by B. Take the upward direction as positive.

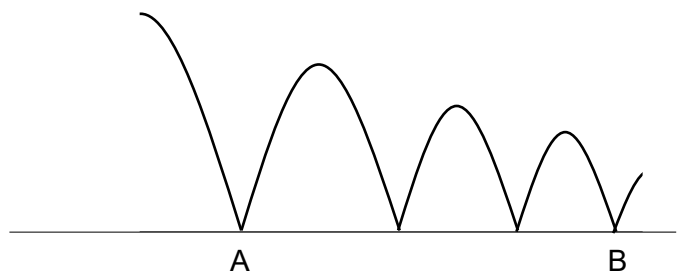


Fig. 1.2

Sketch, on Fig. 1.3, a graph to show the variation with time of the vertical velocity of the ball between from the instant it leaves A to the instant it reaches B.

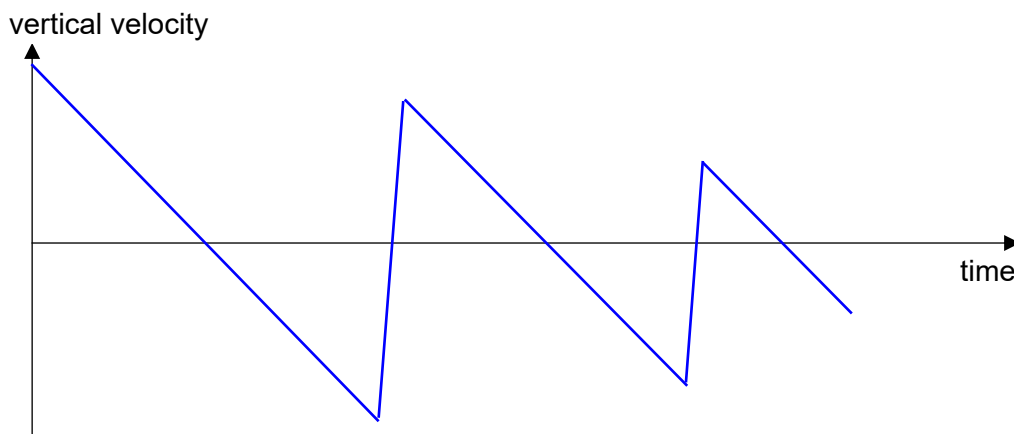


Fig. 1.3

3 parallel lines, same gradient, in between bounces; with vertical lines (or very slight slant) when ball is in contact with ground. [1]
 Speed after rebound is lower than speed before rebound and area above and below graph are equal [1]

[2]

[Total: 7]

